

FUNDACIÓN UNIVERSITARIA KONRAD LORENZ

Redes de Comunicaciones 2

Documentación de Configuración

Topología de Red Completa

PNetLab – EIGRP, OSPF, ACLs, NAT, Zabbix

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1. Mapa General de la Topología

La topología está compuesta por tres secciones principales: la red original (R1, R2, R3), la cadena de ciudades colombianas (Paipa, Medellín, Bogotá, Cúcuta, Manizales, Bucaramanga) y los servicios de monitoreo (Zabbix, Grafana). Se utilizan dos protocolos de enrutamiento dinámico: EIGRP AS 50 entre R1-R2 y en toda la cadena de ciudades, y OSPF Área 0 entre R2-R3, con redistribución mutua en R2.

Red	Segmento	Enlace
192.168.0.0/24	LAN Docker4/Docker5	R1 e0/0 → SW
10.0.12.0/30	Enlace R1-R2	R1 e0/2 ↔ R2 e0/2
10.0.13.0/30	Enlace R1-R3	R1 e0/3 ↔ R3 e0/2
10.0.23.0/30	Enlace R2-R3	R2 e0/1 ↔ R3 e0/1
172.16.200.0/24	LAN Docker6	R2 e0/0
192.168.200.0/24	LAN Servidor	R3 e0/0
192.168.50.0/24	LAN Zabbix Server	R2 e0/3
10.0.34.0/30	Enlace Docker-Paipa	Docker eth2 ↔ Paipa e0/0
192.168.16.0/24	Enlace Paipa-Bucaramanga	Paipa e0/1 ↔ Bucaramanga e0/0
192.168.17.0/24	Enlace Paipa-Medellín	Paipa e0/2 ↔ Medellín e0/0
192.168.18.0/24	Enlace Medellín-Bogotá	Medellín e0/1 ↔ Bogotá e0/0
192.168.19.0/24	Enlace Bogotá-Cúcuta	Bogotá e0/1 ↔ Cúcuta e0/0
192.168.20.0/24	Enlace Cúcuta-Manizales	Cúcuta e0/1 ↔ Manizales e0/0
192.168.21.0/24	Enlace Manizales-Bucaramanga	Manizales e0/1 ↔ Bucaramanga e0/3
192.168.1.0/24	Enlace Bucaramanga-R16	Bucaramanga e0/2 ↔ R16 e0/3
192.168.100.0/24	LAN Docker Bucaramanga	R16 e0/0 → SW
192.168.22.0/24	LAN Docker Medellín	Medellín e0/2 → R21 SW
10.0.45.0/30	Enlace Manizales-Fortinet	Manizales e0/2 ↔ Fortinet e0/0
10.0.46.0/30	Enlace Fortinet-R23	Fortinet e0/1 ↔ R23 e0/0
10.0.137.0/24	NAT Internet	R2 e1/0 (DHCP)

2. Switch SW (Arista – LAN Docker4/Docker5)

```
conf t
hostname SW1

interface ethernet0/0
  switchport mode access
  switchport access vlan 1
  no shutdown

interface ethernet0/1
  switchport mode access
  switchport access vlan 1
  no shutdown

interface ethernet0/2
  switchport mode access
  switchport access vlan 1
  no shutdown

end
write memory
```

3. Router R1

3.1 Interfaces

```
conf t
hostname R1

interface ethernet0/0
  description hacia-SW-LAN
  ip address 192.168.0.1 255.255.255.0
  no shutdown

interface ethernet0/2
  description hacia-R2
  ip address 10.0.12.1 255.255.255.252
  no shutdown

interface ethernet0/3
  description hacia-R3
  ip address 10.0.13.1 255.255.255.252
  no shutdown

end
write memory
```

3.2 EIGRP

```
conf t
```

```
router eigrp 50
 network 192.168.0.0 0.0.0.255
 network 10.0.12.0 0.0.0.3
 network 10.0.13.0 0.0.0.3
 no auto-summary
end
write memory
```

3.3 ACLs

```
conf t
ip access-list extended ACL_LAN_IN
 deny tcp host 192.168.0.50 host 192.168.200.254 eq 80
 deny tcp host 192.168.0.50 host 192.168.200.254 eq 443
 deny icmp host 192.168.0.51 host 192.168.200.254
 deny ip 192.168.0.0 0.0.0.255 172.16.200.0 0.0.0.255
 permit ip any any

interface ethernet0/0
 ip access-group ACL_LAN_IN in

ip access-list standard ACL_TELNET
 permit host 192.168.0.50
 deny any

line vty 0 4
 access-class ACL_TELNET in

end
write memory
```

4. Router R2

4.1 Interfaces

```
conf t
hostname R2

interface ethernet0/0
 description hacia-Docker6
 ip address 172.16.200.1 255.255.255.0
 ip nat inside
 no shutdown

interface ethernet0/1
 description hacia-R3
 ip address 10.0.23.1 255.255.255.252
 ip nat inside
 no shutdown

interface ethernet0/2
 description hacia-R1
```

```

ip address 10.0.12.2 255.255.255.252
ip nat inside
no shutdown

interface ethernet0/3
description hacia-Zabbix-server
ip address 192.168.50.1 255.255.255.0
ip nat inside
no shutdown

interface ethernet1/0
description hacia-Internet
ip address dhcp
ip nat outside
no shutdown

end
write memory

```

4.2 EIGRP + OSPF + Redistribución

```

conf t
router eigrp 50
network 10.0.12.0 0.0.0.3
network 172.16.200.0 0.0.0.255
network 192.168.50.0 0.0.0.255
redistribute ospf 1 metric 1000 100 255 1 1500
redistribute static metric 1000 100 255 1 1500
no auto-summary

router ospf 1
network 10.0.23.0 0.0.0.3 area 0
redistribute eigrp 50 subnets

end
write memory

```

4.3 NAT

```

conf t
ip access-list standard NAT_LIST
permit 192.168.0.0 0.0.255.255
permit 172.16.0.0 0.0.255.255
permit 10.0.0.0 0.255.255.255

ip nat inside source list NAT_LIST interface ethernet1/0 overload
ip route 0.0.0.0 0.0.0.0 10.0.137.1

end
write memory

```

4.4 Telnet ACL

```

conf t

```

```
ip access-list standard ACL_TELNET
  permit host 192.168.0.50
  deny any
```

```
line vty 0 4
  access-class ACL_TELNET in
```

```
end
write memory
```

5. Router R3 (Paipa-R3)

5.1 Interfaces

```
conf t
hostname Paipa-R3

interface ethernet0/0
  description hacia-Servidor
  ip address 192.168.200.1 255.255.255.0
  no shutdown

interface ethernet0/1
  description hacia-R2
  ip address 10.0.23.2 255.255.255.252
  no shutdown

interface ethernet0/2
  description hacia-R1
  ip address 10.0.13.2 255.255.255.252
  no shutdown

interface ethernet0/3
  description hacia-Paipa
  ip address 10.0.34.1 255.255.255.252
  no shutdown

end
write memory
```

5.2 OSPF + EIGRP

```
conf t
router ospf 1
  network 192.168.200.0 0.0.0.255 area 0
  network 10.0.23.0 0.0.0.3 area 0
  network 10.0.13.0 0.0.0.3 area 0

router eigrp 50
  network 10.0.34.0 0.0.0.3

ip route 10.0.34.0 255.255.255.252 192.168.200.254
```

```
end
write memory
```

5.3 ACLs

```
conf t
ip access-list extended ACL_SERVER_IN
  deny icmp host 192.168.200.254 any echo
  permit ip any any

interface ethernet0/0
  ip access-group ACL_SERVER_IN in

ip access-list standard ACL_TELNET
  permit host 192.168.0.50
  deny any

line vty 0 4
  access-class ACL_TELNET in

end
write memory
```

6. Docker Servidor (192.168.200.254)

Este Docker actúa como puente entre R3 y Paipa, con dos interfaces de red (eth1 y eth2) y reenvío de paquetes habilitado.

```
# eth1: 192.168.200.254/24 (hacia R3)
# eth2: 10.0.34.1/30 (hacia Paipa)

sudo ip addr add 10.0.34.1/30 dev eth2
sudo sysctl -w net.ipv4.ip_forward=1
sudo ip route add default via 192.168.200.1
```

7. Router Paipa

```
conf t
hostname Paipa

interface ethernet0/0
  description hacia-Docker-R3
  ip address 10.0.34.2 255.255.255.252
  no shutdown

interface ethernet0/1
  description hacia-Bucaramanga
  ip address 192.168.16.1 255.255.255.0
  no shutdown
```

```
interface ethernet0/2
description hacia-Medellin
ip address 192.168.17.1 255.255.255.0
no shutdown
```

```
router eigrp 50
network 10.0.34.0 0.0.0.3
network 192.168.16.0 0.0.0.255
network 192.168.17.0 0.0.0.255
no auto-summary
```

```
end
write memory
```

8. Router Medellín

```
conf t
hostname Medellin
```

```
interface ethernet0/0
description hacia-Paipa
ip address 192.168.17.2 255.255.255.0
no shutdown
```

```
interface ethernet0/1
description hacia-Bogota
ip address 192.168.18.1 255.255.255.0
no shutdown
```

```
interface ethernet0/2
description LAN-R21-switch
ip address 192.168.22.1 255.255.255.0
no shutdown
```

```
router eigrp 50
network 192.168.17.0 0.0.0.255
network 192.168.18.0 0.0.0.255
network 192.168.22.0 0.0.0.255
no auto-summary
```

```
end
write memory
```

9. Router Bogotá

```
conf t
hostname Bogota
```

```
interface ethernet0/0
```

```
description hacia-Medellin
ip address 192.168.18.2 255.255.255.0
no shutdown

interface ethernet0/1
description hacia-Cucuta
ip address 192.168.19.1 255.255.255.0
no shutdown

router eigrp 50
network 192.168.18.0 0.0.0.255
network 192.168.19.0 0.0.0.255
no auto-summary

end
write memory
```

10. Router Cúcuta

```
conf t
hostname Cucuta

interface ethernet0/0
description hacia-Bogota
ip address 192.168.19.2 255.255.255.0
no shutdown

interface ethernet0/1
description hacia-Manizales
ip address 192.168.20.1 255.255.255.0
no shutdown

router eigrp 50
network 192.168.19.0 0.0.0.255
network 192.168.20.0 0.0.0.255
no auto-summary

end
write memory
```

11. Router Manizales

```
conf t
hostname Manizales

interface ethernet0/0
description hacia-Cucuta
ip address 192.168.20.2 255.255.255.0
no shutdown
```

```

interface ethernet0/1
  description hacia-Bucaramanga
  ip address 192.168.21.1 255.255.255.0
  no shutdown

interface ethernet0/2
  description hacia-Fortinet
  ip address 10.0.45.1 255.255.255.252
  no shutdown

router eigrp 50
  network 192.168.20.0 0.0.0.255
  network 192.168.21.0 0.0.0.255
  network 10.0.45.0 0.0.0.3
  no auto-summary

end
write memory

```

12. Router Bucaramanga

```

conf t
hostname Bucaramanga

interface ethernet0/0
  description hacia-Paipa
  ip address 192.168.16.2 255.255.255.0
  no shutdown

interface ethernet0/1
  description hacia-Manizales
  ip address 192.168.21.2 255.255.255.0
  no shutdown

interface ethernet0/2
  description hacia-R16
  ip address 192.168.1.1 255.255.255.0
  no shutdown

interface ethernet0/3
  description LAN-R17-switch
  ip address 192.168.100.1 255.255.255.0
  no shutdown

router eigrp 50
  network 192.168.16.0 0.0.0.255
  network 192.168.21.0 0.0.0.255
  network 192.168.1.0 0.0.0.255
  network 192.168.100.0 0.0.0.255
  no auto-summary

end

```

```
write memory
```

13. Router R16

```
conf t
hostname R16

interface ethernet0/3
description hacia-Bucaramanga
ip address 192.168.1.2 255.255.255.0
no shutdown

interface ethernet0/0
description LAN-Switch-Docker
ip address 192.168.100.1 255.255.255.0
no shutdown

interface ethernet0/1
description hacia-Cisco
ip address 10.0.0.1 255.255.255.252
no shutdown

router eigrp 50
network 192.168.1.0 0.0.0.255
network 192.168.100.0 0.0.0.255
network 10.0.0.0 0.0.0.3
no auto-summary

end
write memory
```

14. Fortinet (simulado con IOL)

```
conf t
hostname Fortinet

interface ethernet0/0
description hacia-Manizales
ip address 10.0.45.2 255.255.255.252
no shutdown

interface ethernet0/1
description hacia-R23
ip address 10.0.46.1 255.255.255.252
no shutdown

ip access-list extended FIREWALL
permit icmp any any
permit tcp any any eq 80
permit tcp any any eq 443
```

```
permit tcp any any eq 22
deny ip any any

router eigrp 50
 network 10.0.45.0 0.0.0.3
 network 10.0.46.0 0.0.0.3
 no auto-summary

end
write memory
```

15. Switches Secundarios

15.1 R17-SW (Bucaramanga)

```
conf t
hostname R17-SW
interface ethernet0/0
 switchport mode access
 no shutdown
interface ethernet0/1
 switchport mode access
 no shutdown
end
write memory
```

15.2 R18-SW

```
conf t
hostname R18-SW
interface ethernet0/0
 switchport mode access
 no shutdown
end
write memory
```

15.3 R21-SW (Medellín)

```
conf t
hostname R21-SW
interface ethernet0/0
 switchport mode access
 no shutdown
interface ethernet0/1
 switchport mode access
 no shutdown
end
write memory
```

15.4 R23-SW (Fortinet)

```
conf t
hostname R23-SW
```

```
interface ethernet0/0
  switchport mode access
  no shutdown
end
write memory
```

16. Configuración de Contenedores Docker

16.1 Docker4 (192.168.0.50)

```
ip addr add 192.168.0.50/24 dev eth1
ip route add default via 192.168.0.1
```

16.2 Docker5 (192.168.0.51)

```
ip addr add 192.168.0.51/24 dev eth1
ip route add default via 192.168.0.1
```

16.3 Docker6 (172.16.200.2)

```
ip addr add 172.16.200.2/24 dev eth1
ip route add default via 172.16.200.1
```

16.4 Docker Medellín (192.168.22.10)

```
ip addr add 192.168.22.10/24 dev eth1
ip route add default via 192.168.22.1
```

16.5 Docker Bucaramanga (192.168.100.10)

```
ip addr add 192.168.100.10/24 dev eth1
ip route add default via 192.168.100.1
```

16.6 Docker Zabbix-Server (192.168.50.10)

```
ip addr add 192.168.50.10/24 dev eth1
ip route add 192.168.0.0/16 via 192.168.50.1 dev eth1
ip route add 172.16.0.0/16 via 192.168.50.1 dev eth1
ip route add 10.0.0.0/8 via 192.168.50.1 dev eth1
```

17. Resumen de ACLs

Requisito	Router	ACL	Acción
Req 1: .50 no accede web al servidor	R1	ACL_LAN_IN	Deny TCP 80/443 de .50 a servidor
Req 2: .51 no hace ping al servidor	R1	ACL_LAN_IN	Deny ICMP de .51 a servidor

Req 3: Servidor no inicia ping	R3	ACL_SERVER_IN	Deny ICMP echo del servidor
Req 4: Red .0/24 no accede a 172.16	R1	ACL_LAN_IN	Deny IP red .0/24 a 172.16.200.0/24
Req 5: Solo .50 hace telnet	R1/R2/R3	ACL_TELNET	Permit solo host .50 en VTY

18. Resumen de Protocolos de Enrutamiento

Protocolo	AS/Área	Routers	Redes
EIGRP	AS 50	R1, R2	192.168.0.0, 10.0.12.0, 10.0.13.0, 172.16.200.0, 192.168.50.0
OSPF	Área 0	R2, R3	10.0.23.0, 192.168.200.0, 10.0.13.0
Redistribución	R2	EIGRP ↔ OSPF	Mutua en R2
EIGRP	AS 50	Paipa-Bucaramanga	Toda la cadena de ciudades
NAT	R2 e1/0	Overload	Salida a Internet

19. Instalación de Zabbix Server

Zabbix 5.0 instalado directamente en el servidor PNetLab (192.168.183.128) con MySQL 5.7.

19.1 Base de Datos

```
mysql -e "CREATE DATABASE zabbix CHARACTER SET utf8 COLLATE utf8_bin;"
mysql -e "CREATE USER 'zabbix'@'localhost' IDENTIFIED BY 'zabbix123';"
mysql -e "GRANT ALL PRIVILEGES ON zabbix.* TO 'zabbix'@'localhost';"
mysql -e "FLUSH PRIVILEGES;"
zcat /usr/share/doc/zabbix-server-mysql*/create.sql.gz | mysql -uzabbix -pzabbix123 zabbix
```

19.2 Configuración

```
sed -i 's/# DBPassword=DBPassword=zabbix123/' /etc/zabbix/zabbix_server.conf
sed -i "s/;date.timezone =/date.timezone = America\Bogota/" /etc/php/7.2/apache2/php.ini
```

19.3 Inicio de Servicios

```
service apache2 start
zabbix_server -c /etc/zabbix/zabbix_server.conf
service zabbix-agent start
```

19.4 Acceso Web

Servicio	URL	Usuario	Contraseña
Zabbix	http://192.168.183.128/zabbix	Admin	zabbix

20. Instalación de Zabbix Agent en Dockers

```
apt update
apt install -y zabbix-agent
echo "Server=192.168.183.128" >> /etc/zabbix/zabbix_agentd.conf
echo "ServerActive=192.168.183.128" >> /etc/zabbix/zabbix_agentd.conf
echo "Hostname=NOMBRE_HOST" >> /etc/zabbix/zabbix_agentd.conf
service zabbix-agent start
```

Host	IP	Hostname en Zabbix
Docker4	192.168.0.50	Docker4
Docker5	192.168.0.51	Docker5
Docker6	172.16.200.2	Docker6
Servidor	192.168.200.254	ServidorWeb
Zabbix Docker	192.168.50.10	Zabbix-Server

21. Script de Inicio Rápido

Ejecutar en MobaXterm cada vez que se encienda PNetLab:

```
#!/bin/bash
# /root/iniciar_zabbix.sh
mkdir -p /var/run/mysqld
chown mysql:mysql /var/run/mysqld
service mysql start
service apache2 start
zabbix_server -c /etc/zabbix/zabbix_server.conf
service zabbix-agent start
echo "Zabbix listo en http://192.168.183.128/zabbix"
```

22. Pruebas de Verificación de ACLs

Desde	Hacia	Qué	Resultado
Docker5 (.51)	192.168.200.254	Ping	BLOQUEADO
Docker4 (.50)	192.168.200.254	Ping	FUNCIONA
Docker4 (.50)	192.168.200.254	HTTP	BLOQUEADO
Servidor	192.168.0.50	Ping	BLOQUEADO

Docker4 (.50)	172.16.200.2	Ping	BLOQUEADO
Docker5 (.51)	172.16.200.2	Ping	BLOQUEADO
Docker4 (.50)	R1/R2/R3	Telnet	FUNCIONA
Docker5 (.51)	R1/R2/R3	Telnet	BLOQUEADO

23. Comandos de Verificación

! Ver interfaces y estado
show ip interface brief

! Ver tabla de rutas
show ip route

! Ver vecinos EIGRP
show ip eigrp neighbors
show ip eigrp topology

! Ver vecinos OSPF
show ip ospf neighbor
show ip ospf database

! Ver ACLs y hits
show ip access-lists

! Ver NAT
show ip nat translations

! Ver configuración completa
show running-config